

Final Examination

December 29, 2021

Please use the Google form titled **Final Exam - Responses** to answer the questions.

If you need to upload solutions to some problems, upload a single file with solutions to all the problems for which you need to provide written solutions and name the file as per the following convention: *YourName_RollNumber_FE*. Thus for example, if Paul Erdos, with Roll Number 1911001, wanted to upload the solution to some problems of the current quiz, he would have to name the file containing all the solutions as *PaulErdos_1911001_FE*.

2 points for following the naming convention and uploading in the correct place (using the Google form). **Max points = 40.**

Problem 1 (8 points). Determine the truth values of the following statements assuming that the domain for all the variables is the set of all real numbers.

- (i) $\forall x (x < 0) \longrightarrow \exists y \forall z (y > z)$
- (ii) $\exists x (x < 0) \longrightarrow \forall y \exists z (y > z)$
- (iii) $\forall x \exists y (x \leq 0 \wedge xy < 0)$
- (iv) $\forall x \forall y (x \leq y \longrightarrow \exists z (x \leq z \wedge z \leq y))$

Problem 2 (5 points). Suppose R and S are binary relations on a set $A \neq \emptyset$. Show that $(R \circ S)^{-1} = S^{-1} \circ R^{-1}$.

Problem 3 (5 points). Suppose X is a finite set. Show that a function $f : X \rightarrow X$ is injective iff it is surjective. Give an example to show that this is not the case if X is infinite.

Problem 4 (5 points). Let p be a prime and let $k \geq 1$ be a natural number. Prove that for $k < p$, $\binom{p}{k}$ is divisible by p .

Problem 5 (5 points). Find the coefficient of x^2y^8z in $(2x + y^2 - 5z)^7$.

Problem 6 (5 points). Is it true that if two graphs G and H are isomorphic, then there exists a bijection $f : E(G) \rightarrow E(H)$? Justify your answer.

Problem 7 (5 points). Can you draw a graph with score $(1, 2, 2, 3)$? Justify your answer. If yes, then use the Scores Theorem to draw such a graph.